Separating Reconstruction Scale and Label Semantics in Semi-Supervised Nonnegative Matrix Factorization

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Abstract

Semi-supervised nonnegative matrix factorization (NMF) commonly uses the same latent coefficient for sample reconstruction and class representation, causing one-hot label constraints to fix both semantic direction and reconstruction amplitude. We propose scale-semantic separated NMF (ScaleSem-NMF), which decomposes each coefficient into a nonnegative amplitude and a simplex-constrained semantic direction. Label constraints and graph propagation operate on the semantic directions, while reconstruction determines the amplitudes. The resulting model integrates reconstruction fidelity with positive-semidefinite high-order diffusion. We derive a multiplicative basis update, an exact amplitude update, and an Armijo projected-gradient semantic update. Experiments on eight datasets with 20 paired seeds demonstrate substantial ACC and NMI improvements on COIL20 and COIL100 and consistent gains on AR and MNIST relative to HLPNMF. Experiments across five supervision levels confirm the effectiveness of scale-semantic separation under limited supervision. Noise robustness, convergence, runtime, and parameter analyses further establish the stability, optimization behavior, and computational characteristics of the proposed method. The source code is publicly available at https://github.com/matrix-research-lab-Wf/ScaleSem-NMF.

Keywords: semi-supervised clustering, nonnegative matrix factorization, scale-semantic separation, label propagation, simplex representation

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1. Introduction

Clustering has been surveyed by Jain et al. [1] and Jain [2], while semi-supervised learning and clustering are reviewed by van Engelen and Hoos [3] and Bair [4]. Nonnegative matrix factorization (NMF) provides additive latent representations [5, 6]; Gillis [7] discusses its modeling and optimization properties. Constrained NMF incorporates labels [8], while discriminative variants refine class representations [9–11]. Graph NMF models local geometry [12], and hypergraph NMF extends it to group-wise structure [13]. Label propagation enters NMF through the methods of Yi et al. [14], Lan et al. [15], and Wang et al. [16].

Compact label-guided NMF models frequently use the same coefficient vector for two roles. In the reconstruction model, its magnitude controls sample amplitude. In the supervision model, a one-hot vector represents normalized class semantics. Directly setting a labeled coefficient equal to its one-hot label therefore fixes both its direction and its radial magnitude. The reconstruction amplitude remains sample and basis dependent even after input normalization.

Direct and propagation-guided models provide the closest context. GOC-NMF and OCSNMF impose one-hot semantics directly in graph or symmetric factorizations [17, 18]; constraint propagation [19], hypergraph propagation [20], pivotal-aware propagation [21], and joint label/constraint propagation [22] transfer supervision through different relational objects. Robust semi-supervised NMF further changes the reconstruction loss [23], whereas magnitude-aware supervision explicitly questions the information discarded

by binary labels [24]. SNMFWLP and HLPNMF couple reconstruction with weighted or high-order propagation through learned representations [25, 26]. These methods motivate a precise question: can reconstruction amplitude and class semantics be represented separately while preserving a compact, directly decodable NMF model?

ScaleSem-NMF answers this question by writing every coefficient as

$$v_i = s_i f_i, \qquad s_i \geq 0, \quad f_i \geq 0, \quad \mathbf{1}_c^\top f_i = 1.$$

The scalar s_i is reconstruction driven, whereas f_i is a semantic direction. Only f_i is clamped to a known label and smoothed over the graph. Section 2 discusses magnitude-aware supervision and simplex-structured factorization; ScaleSem-NMF jointly learns sample amplitudes while propagating semantic directions. The contributions are fourfold.

1. We identify point anchoring as a structural coupling between semantic direction and reconstruction amplitude and replace it with a sample-wise scale-semantic decomposition.
2. We formulate a compact ScaleSem-NMF objective with a simplex semantic matrix and a symmetric positive-semidefinite high-order graph operator.
3. We give block updates with an exact amplitude step and feasibility-preserving semantic descent, together with explicit assumptions for monotonicity and conditional stationarity.
4. We evaluate the model under protocols fixed before formal analysis. The evidence includes eight-dataset paired comparisons, five supervision budgets, parameter sensitivity, controlled ablations, salt-and-pepper

corruption, incorrect-label corruption, convergence, unified single-thread timing, and nested PIE resource measurements.

2. Related work

2.1. Semi-supervised NMF

Standard NMF approximates $X \in \mathbb{R}_+^{d\times n}$ by $UV^\top$, where $U \in \mathbb{R}_+^{d\times c}$ is a basis and $V \in \mathbb{R}_+^{n\times c}$ contains sample coefficients. Semi-supervised variants constrain V [8], learn discriminative representations [9], regularize sample graphs [12], or integrate label propagation [14]. ScaleSem-NMF uses direct class prediction from the largest semantic coordinate. S4NMF derives self-supervision from the consensus of ensembled NMF solutions and feeds similarity and dissimilarity information back into factorization [27]. ScaleSem-NMF addresses the coefficient geometry of direct label anchoring through a sample-wise amplitude and semantic direction.

2.2. Label propagation and high-order diffusion

Let $A = A^\top \geq 0$ be a symmetric sample affinity matrix and $D = \mathrm{diag}(A\mathbf{1}_n)$, with strictly positive degrees. The normalized symmetric operator is $P = D^{-1/2}AD^{-1/2}$. Graph label propagation assumes that nearby samples have similar semantic representations [28, 29]. Graph construction and hypergraph learning are reviewed in [30, 31]; Song et al. [32] survey graph-based semi-supervised learning. HLPNMF uses multi-step propagation inside NMF [26]. ScaleSem-NMF retains propagation depth but applies it only to the normalized semantic direction.

2.3. Magnitude-aware and simplex representations

Magnitude-aware supervision observes that binary labels may discard useful sample scale [24]. Simplex factorizations interpret normalized coefficients as soft memberships [33], while dual-simplex formulations use polytope polarity to control identifiability and volume [34]. Discriminative graph NMF provides a related example of coupling neighborhood structure with label separation [35]. ScaleSem-NMF extends these geometric insights to partially labeled graph propagation by combining a learned radial amplitude with a simplex semantic direction, so that reconstruction and supervision act on different variables.

3. Methodology

3.1. Point anchoring and scale-semantic decomposition

For labeled sample i, direct coefficient anchoring imposes $v_i = y_i$, where y_i is one-hot. Under $x_i \approx U v_i^\top$, this forces $x_i \approx U y_i^\top$: every labeled sample in a class is assigned the same unit latent amplitude. ScaleSem-NMF instead defines

$$v_i = s_i f_i, \qquad f_i \in \Delta^{c-1} = \{f \in \mathbb{R}_+^c : \mathbf{1}_c^\top f = 1\}.$$

With $F = [f_1^\top; \ldots; f_n^\top] \in \mathbb{R}_+^{n \times c}$ and $s \in \mathbb{R}_+^n$, $V = \mathrm{diag}(s)F$. For any nonzero v_i, the decomposition is uniquely fixed by $s_i = \|v_i\|_1$ and $f_i = v_i / s_i$. A global gauge between U and s remains and is normalized numerically.

This decomposition alone is a row-wise reparameterization of a nonnegative coefficient matrix. Indeed, if every row of $V \geq 0$ has positive sum and $r = V\mathbf{1}_c$, then

$$s = r, \qquad F = \mathrm{diag}(r)^{-1} V, \qquad V = \mathrm{diag}(s)F. \tag{1}$$

Consequently, the reconstruction term is unchanged. The substantive modeling choice is that labels and high-order diffusion act on the row-normalized semantic directions F, rather than on the amplitude-bearing coefficients V. Equivalently, over positive-row V, the regularizer is

$$\alpha \operatorname{Tr}\Big[\Big(\operatorname{diag}(V\mathbf{1}_c)^{-1}V\Big)^{\top} L_k \Big(\operatorname{diag}(V\mathbf{1}_c)^{-1}V\Big)\Big]. \tag{2}$$

If $s_i = 0$, reconstruction is independent of f_i; its semantic direction remains constrained by the simplex, the labeled rows, and graph diffusion. The stationarity result below therefore states explicit nondegeneracy conditions for the reconstruction blocks.

3.2. Objective and graph operator

Figure 1 summarizes the role-specific coupling. Its upper branch constructs the positive-semidefinite high-order diffusion operator from sample geometry; its lower branch decomposes each coefficient into reconstruction amplitude s_i and simplex semantic direction f_i. Reconstruction uses U, s, F, whereas partial labels and diffusion constrain only F. Prediction is decoded directly from F, and propagation acts on semantic directions.

All experiments use an affinity induced by a weighted neighborhood hypergraph. For samples x_i, define the Gaussian similarities

$$K_{ij} = \exp\left(-\frac{\|x_i - x_j\|_2^2}{\sigma^2}\right), \qquad \sigma = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\|x_i - x_j\|_2. \tag{3}$$

The hyperedge e_i contains center sample i and its p nearest neighbors. With $m = n$ hyperedges, let $H \in \{0,1\}^{m\times n}$ satisfy $H_{ij} = 1$ if sample $j \in e_i$, and

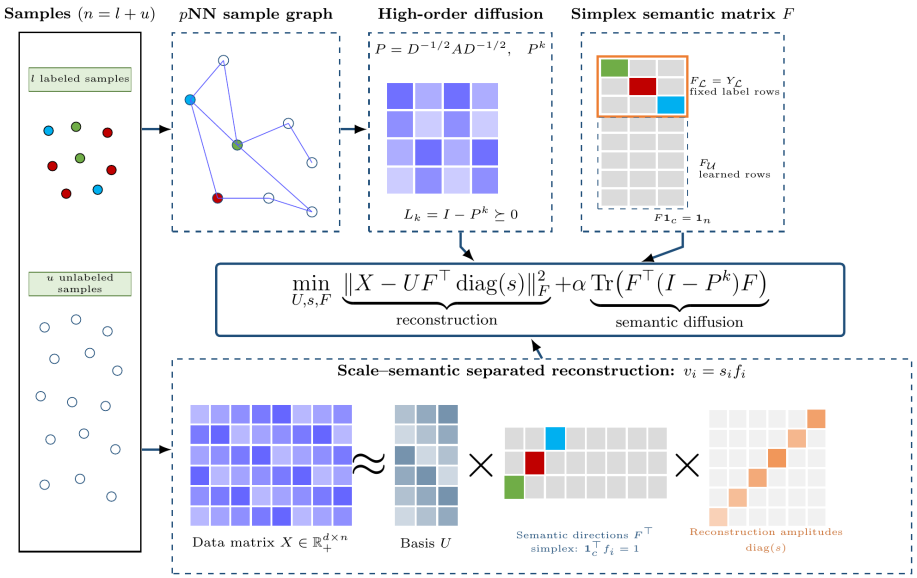


Figure 1: Mechanism of ScaleSem-NMF. The model combines high-order diffusion with scale-semantic coefficient decomposition, applying supervision to the simplex semantic matrix F.

define

$$w_i = \sum_{j \in e_i} K_{ij}, \qquad W = \mathrm{diag}(w_1, \ldots, w_m),$$
$$D_e = \mathrm{diag}(H\mathbf{1}_n), \quad \Theta = H^\top W D_e^{-1} H. \tag{4}$$

The implementation then uses

$$A = \frac{\Theta + \Theta^\top}{2}, \qquad D = \mathrm{diag}(A\mathbf{1}_n), \qquad P = D^{-1/2} A D^{-1/2}. \tag{5}$$

Thus the hypergraph is used to construct a symmetric nonnegative sample affinity; the optimized variables remain U, s, F. For an integer $k \geq 1$, define

$$L_k = I_n - P^k.$$

Proposition 1. If A is symmetric, nonnegative, and has positive degrees, then $L_k \succeq 0$ for every positive integer k.

Proof. The matrix P is symmetric and $P = D^{1/2}(D^{-1}A)D^{-1/2}$, so it is similar to the row-stochastic matrix $D^{-1}A$. Hence the eigenvalues of P are real and belong to $[-1, 1]$. If $\lambda_i(P)$ is an eigenvalue of P, the corresponding

eigenvalue of L_k is $1 - \lambda_i(P)^k \geq 0$ for every positive integer k. Thus $L_k \succeq 0$. □

Let $\mathcal{L}$ be the labeled index set and $Y_{\mathcal{L}}$ the one-hot labels. ScaleSem-NMF solves

$$\begin{aligned} \min_{U,s,F} \quad & J(U,s,F) = \left\| X - UF^\top \operatorname{diag}(s) \right\|_F^2 + \alpha \operatorname{Tr}(F^\top L_k F), \\ \text{s.t.} \quad & U \geq 0, \quad s \geq 0, \quad F \geq 0, \\ & F\mathbf{1}_c = \mathbf{1}_n, \quad F_{\mathcal{L}} = Y_{\mathcal{L}}. \end{aligned} \tag{6}$$

Prediction is direct: $\widehat{y}_i = \arg\max_q F_{iq}$.

3.3. Block optimization

With $V^t = \operatorname{diag}(s^t)F^t$, the basis update is

$$\widetilde{U}^{t+1} = U^t \odot \frac{XV^t}{U^t((V^t)^\top V^t)}. \tag{7}$$

The implementation applies a machine-scale denominator safeguard when required. Theorem 2 assumes that the safeguard is eventually inactive or vanishes, so the limiting update corresponds to the unperturbed objective.

For fixed $\widetilde{U}^{t+1}, F^t$, set $g_i^{t+1} = \widetilde{U}^{t+1}(f_i^t)^\top$. Each amplitude has the exact update

$$\widetilde{s}_i^{t+1} = \begin{cases} \max\left(0, \dfrac{(g_i^{t+1})^\top x_i}{(g_i^{t+1})^\top g_i^{t+1}}\right), & g_i^{t+1} \neq 0, \\ 0, & g_i^{t+1} = 0. \end{cases} \tag{8}$$

The global gauge is normalized by

$$\beta^{t+1} = \|\widetilde{U}^{t+1}\|_F, \qquad U^{t+1} = \frac{\widetilde{U}^{t+1}}{\beta^{t+1}}, \qquad s^{t+1} = \beta^{t+1}\widetilde{s}^{t+1}, \tag{9}$$

which leaves the reconstruction unchanged. Here tildes denote the basis and amplitude after their block updates but before gauge normalization.

Let $D_s = \mathrm{diag}(s)$. The semantic gradient is

$$\nabla_F J = 2D_s^2 F(U^\top U) - 2D_s X^\top U + 2\alpha L_k F. \tag{10}$$

For a backtracking step size $\eta_t > 0$, the semantic update is

$$F^{t+1} = \Pi_{\mathcal{C}}\Big(F^t - \eta_t \nabla_F J(U^{t+1}, s^{t+1}, F^t)\Big), \tag{11}$$

where $\mathcal{C}$ enforces unlabeled simplex rows and fixed labeled rows. Backtracking accepts the candidate $\widehat{F}$ when

$$J(U, s, \widehat{F}) \le J(U, s, F) + \sigma\langle \nabla_F J, \widehat{F} - F\rangle, \qquad 0 < \sigma < 1. \tag{12}$$

For completeness, let $z \in \mathbb{R}^c$ and write its sorted entries as $z_{(1)} \ge \cdots \ge z_{(c)}$. Define

$$q = \max\left\{ j : \; z_{(j)} - \frac{\sum_{\ell=1}^{j} z_{(\ell)} - 1}{j} > 0 \right\}, \qquad \tau = \frac{\sum_{\ell=1}^{q} z_{(\ell)} - 1}{q}. \tag{13}$$

The Euclidean simplex projection is $[\Pi_{\Delta^{c-1}}(z)]_j = \max\{z_j - \tau, 0\}$. Accordingly, $\Pi_{\mathcal{C}}$ applies this projection row-wise to unlabeled samples and restores every labeled row to $Y_{\mathcal{L}}$. This operation costs $O(nc \log c)$ with sorting.

3.4. Algorithm

The complete optimization procedure is summarized in Algorithm 1. Its order is important: the amplitude is minimized after the basis update, gauge normalization is applied before semantic descent, and the stopping test is evaluated only after a complete block cycle.

3.5. Convergence statement

Theorem 1 (monotonic descent and objective convergence). Suppose $L_k \succeq 0$, all multiplicative-update denominators are positive, the denominator safeguard is inactive (or vanishes), and the Armijo search terminates.

Algorithm 1 Optimization procedure for ScaleSem-NMF

Input: Data matrix $X \in \mathbb{R}_+^{d\times n}$; labeled index set $\mathcal{L}$ and one-hot label matrix $Y_{\mathcal{L}}$; number of classes c; neighborhood size p; propagation order k; regularization parameter α; maximum iterations T; tolerance ε; Armijo parameters η_0, ρ, σ.
Output: Basis U, amplitudes s, semantic matrix F, and predicted labels $\widehat{y}$.
Precomputation:

1. Normalize each sample to unit ℓ_2 norm. Construct $K, H, W, D_e, \Theta, A, D, P$ using (3)–(5), and set $L_k = I - P^k$.

Initialization:

2. Draw U_{ij} and an auxiliary coefficient matrix $(V_0)_{ij}$ independently from $\mathrm{Unif}(0,1)$, replace numerical zeros by machine epsilon, set $s = V_0\mathbf{1}_c$ and $F = \mathrm{diag}(s)^{-1}V_0$, and set $F_{\mathcal{L}} = Y_{\mathcal{L}}$. Re-estimate each labeled amplitude using (8) before the first complete iteration.
3. Evaluate $J^{(0)}$ from (6).

Iterative updates:

4. **for** $t = 0, \ldots, T-1$ **do**
   a. Set $V^t = \mathrm{diag}(s^t)F^t$, and compute $\widetilde{U}^{t+1}$ using (7).
   b. For $i = 1, \ldots, n$, set $g_i^{t+1} = \widetilde{U}^{t+1}(f_i^t)^\top$ and compute $\widetilde{s}_i^{t+1}$ using (8).
   c. Normalize $(\widetilde{U}^{t+1}, \widetilde{s}^{t+1})$ using (9).
   d. Compute $\nabla_F J(U^{t+1}, s^{t+1}, F^t)$ using (10). Starting from $\eta_t = \eta_0$, reduce $\eta_t \leftarrow \rho\eta_t$ until the candidate in (11) satisfies (12), and then accept F^{t+1}.
   e. Evaluate $J^{(t+1)}$. If
   $$\frac{|J^{(t+1)} - J^{(t)}|}{\max\{1, J^{(t)}\}} \le \varepsilon,$$
   break.
5. **end for**

Prediction: For every unlabeled sample, assign $\widehat{y}_i = \arg\max_{1\le q\le c} F_{iq}$, and return $U, s, F, \widehat{y}$.

Then every complete ScaleSem-NMF cycle satisfies

$$J(z^{t+1}) \le J(z^t),$$

and the scalar sequence $\{J(z^t)\}$ converges.

Proof. For fixed (s, F), the update of U is the standard Euclidean-NMF multiplicative step and minimizes its auxiliary function, so it cannot increase the reconstruction term. For fixed (U, F), each update of s_i is the exact minimizer of a nonnegative scalar quadratic. The transformation $(U, s) \mapsto (U/\beta, \beta s)$ leaves $UF^\top \mathrm{diag}(s)$, and hence J, unchanged. Finally, the accepted Armijo step decreases or preserves J over the feasible set $\mathcal{C}$. Combining the four block operations gives the stated descent inequality. Since $L_k \succeq 0$, both terms of J are nonnegative; therefore the monotone sequence $\{J(z^t)\}$ has a finite limit. □

Theorem 2 (conditional accumulation-point stationarity). Let $z^t = (U^t, s^t, F^t)$ be the gauge-normalized iterates, with $\|U^t\|_F = 1$. In addition to the conditions of Theorem 1, assume:

(A1) $\{z^t\}$ is contained in a compact set.

(A2) Along every convergent subsequence, the entries of U^t and the denominators of the safeguard-free multiplicative map are bounded away from zero.

(A3) The Euclidean-NMF auxiliary functions and all amplitude subproblems have uniformly positive curvature on the normalized compact iterate set; the gauge normalization and its inverse are uniformly Lipschitz there.

(A4) The accepted projected-Armijo step sizes satisfy $0 < \eta_{\min} \le \eta_t \le \eta_{\max} < \infty$.

Then every accumulation point $z^* = (U^*, s^*, F^*)$ satisfies

$$\nabla_U J(z^*) = 0, \qquad 0 \le s^* \perp \nabla_s J(z^*) \ge 0, \qquad 0 \in \nabla_F J(z^*) + N_{\mathcal{C}}(F^*),$$

where $N_{\mathcal{C}}$ is the normal cone of the labeled-row/simplex feasible set. Thus every accumulation point is a first-order stationary point of the constrained objective, modulo the objective-invariant global (U, s) gauge.

Lemma 1 (uniform sufficient decrease). Under (A2)-(A4), there exist constants $c_{Us}, c_F > 0$, independent of t, such that the combined basis-amplitude block and the semantic block satisfy

$$J(z^t) - J(z^{t,s}) \ge c_{Us}\Big(\|U^{t+1} - U^t\|_F^2 + \|s^{t+1} - s^t\|_2^2\Big),$$
$$J(z^{t,s}) - J(z^{t+1}) \ge c_F \|F^{t+1} - F^t\|_F^2,$$

where $z^{t,s}$ denotes the intermediate iterate after the amplitude–gauge block.

Proof. For the U-block, the Euclidean-NMF auxiliary function majorizes

the reconstruction term, is tight at U^t, and, by (A3), has Hessian bounded below by a positive constant. Its minimizer therefore yields a quadratic decrease bound. The amplitude objective is a sum of independent scalar quadratics with second derivatives $2\|g_i^{t+1}\|_2^2$; their uniform positive lower bound in (A3) gives another quadratic decrease bound. Adding these two bounds controls the joint unnormalized block displacement. Gauge normalization leaves the objective unchanged, and the uniform bi-Lipschitz property in (A3) transfers this joint bound to $(U^{t+1}, s^{t+1}) - (U^t, s^t)$, proving the first inequality. For the projected semantic step, the projection variational inequality gives

$$\langle \nabla_F J, F^{t+1} - F^t \rangle \leq -\eta_t^{-1} \|F^{t+1} - F^t\|_F^2.$$

Combining this inequality with (12) and (A4) gives the third bound with $c_F = \sigma/\eta_{\max}$. □

Proof of Theorem 2. Summing the two inequalities in Lemma 1 and setting $c = \min\{c_{Us}, c_F\} > 0$ gives

$$J(z^t) - J(z^{t+1}) \geq c\Big(\|U^{t+1} - U^t\|_F^2 + \|s^{t+1} - s^t\|_2^2 + \|F^{t+1} - F^t\|_F^2\Big).$$

Theorem 1 and telescoping summation imply that the three block displacements vanish. By (A1), every convergent subsequence has a one-step-shifted subsequence with the same limit z^*.

Let $\mathcal{M}_U(z^*)$ denote the safeguard-free multiplicative basis update before gauge normalization, and set $\beta^* = \|\mathcal{M}_U(z^*)\|_F$. Continuity and vanishing normalized displacement give $\mathcal{M}_U(z^*) = \beta^* U^*$. Writing $V^* = \mathrm{diag}(s^*)F^*$, positivity of U^* permits elementwise cancellation in the multiplicative map and yields

$$XV^* = \beta^* U^*((V^*)^\top V^*).$$

Exact amplitude minimization and continuity give $0 \le s^* \perp \nabla_s J(z^*) \ge 0$. The objective is invariant under $(U, s) \mapsto (\gamma U, s/\gamma)$; differentiating this identity at $\gamma = 1$ gives

$$\langle \nabla_U J(z^*), U^* \rangle - \langle \nabla_s J(z^*), s^* \rangle = 0.$$

The amplitude complementarity condition makes the second inner product zero. On the other hand,

$$\nabla_U J(z^*) = 2(1 - \beta^*) U^* ((V^*)^\top V^*),$$

and hence

$$0 = \langle \nabla_U J(z^*), U^* \rangle = 2(1 - \beta^*) \| U^* (V^*)^\top \|_F^2.$$

The denominator condition in (A2) makes the final norm positive, so $\beta^* = 1$ and $\nabla_U J(z^*) = 0$. Finally, vanishing projected displacement and (A4) give, along a further step-size subsequence,

$$F^* = \Pi_{\mathcal{C}}(F^* - \eta^* \nabla_F J(z^*)) \quad \Longleftrightarrow \quad 0 \in \nabla_F J(z^*) + N_{\mathcal{C}}(F^*).$$

These are the block KKT conditions. □

3.6. Complexity

In the dense implementation, constructing all pairwise distances and Gaussian similarities costs $O(dn^2)$. Sorting the distances to form n neighborhood hyperedges costs $O(n^2 \log n)$, and forming the dense normalized affinity requires $O(n^2)$ storage. Direct multiplication for P^k costs $O(kn^3)$ in the worst case. Once L_k has been constructed, one optimization iteration costs

$$O(dnc + nc^2 + \mathrm{nnz}(L_k)c)$$

plus $O(nc \log c)$ for sorting-based simplex projection. For T iterations, the total dense time complexity is

$$O\Big(dn^2 + n^2 \log n + kn^3 + T\{dnc + nc^2 + \mathrm{nnz}(L_k)c + nc \log c\}\Big) .$$

The retained matrices require $O(n^2 + dn + dc + nc)$ memory. The n^2 graph term dominates storage when P^k becomes dense.

4. Experiments

4.1. Datasets and protocol

Eight nonnegative datasets were used, with their characteristics summarized in Table 1. Except for Optdigits, the benchmark datasets follow the versions distributed with the implementation of HLPNMF [26]. Optdigits follows the standard 5620-sample UCI benchmark version. Every sample was converted to double precision and normalized to unit Euclidean norm, and class labels were remapped to consecutive integers. The MNIST experiments use the 6996-sample version supplied with the benchmark code.

Table 1: Characteristics of the benchmark datasets.

Dataset	Samples	Features	Classes
YaleB	2414	1024	38
COIL20	1440	1024	20
Optdigits	5620	64	10
PIE	2856	1024	68
AR	1680	2000	120
MSRA25	1799	256	12
COIL100	7200	1024	100
MNIST	6996	784	10

4.2. Compared methods and implementation environment

ScaleSem-NMF was compared with HLPNMF, GOCNMF, and SNM-FWLP. HLPNMF combines constrained NMF with high-order hypergraph

label propagation and assigns a class by the largest coefficient coordinate [26]. GOCNMF uses graph regularization with one-hot coefficient constraints and the native largest-coordinate decision rule [17]. SNMFWLP learns a weighted label-propagation representation and applies its native 50-restart K-means decision procedure [25]. Each adapter preserves the baseline's graph construction, optimizer, stopping rule, and decision mechanism. All methods received the same normalized samples, labeled indices, and supervision budget. Experiments were run in MATLAB R2019a on Windows 10 with an Intel Core i5-8300H CPU and 7.9 GB RAM, without GPU acceleration.

Table 2 gives the ScaleSem-NMF parameters fixed for the main comparison. External baseline parameters were fixed in the corresponding adapters before formal analysis. GOCNMF uses dataset-specific neighborhood sizes and regularization weights, while SNMFWLP uses $\alpha = 10^3$, $\beta = 10^5$, and dataset-specific neighborhood sizes. The executable configuration files record every dataset-level value.

Table 2: Parameters for the main comparison. ScaleSem-NMF and HLPNMF use $\alpha = 10^3$ and the listed p, k; their iteration limits are 200 and 30, respectively. SNMFWLP uses $\alpha = 10^3$ and $\beta = 10^5$.

Dataset	ScaleSem/HL p	ScaleSem/HL k	GOCNMF p	GOCNMF α	SNMFWLP p
YaleB	2	3	2	1000	3
COIL20	2	3	2	10	3
Optdigits	5	1	4	10	4
PIE	3	5	3	1000	3
AR	5	1	3	1000	3
MSRA25	10	1	3	1000	3
COIL100	2	3	2	1000	3
MNIST	5	1	5	100	4

The main comparison used class-stratified 10% supervision, identical labeled splits across methods, and evaluation on unlabeled samples. Each

class contributed $\max\{1, \lfloor 0.1 n_q \rfloor\}$ labeled samples. ScaleSem-NMF initialized U and V_0 uniformly and used a relative-objective tolerance of 10^{-7}, with the stopping test enabled after 30 complete iterations. A 10^{-12} denominator safeguard and an Armijo line search with $\sigma = 10^{-4}$ were applied. Baselines retained their native solvers and decision rules, including the 50-restart K-means procedure of SNMFWLP. Six clustering metrics were evaluated under the same paired protocol. Optimal assignment was applied only to ACC [36], while NMI and paired comparisons followed [37, 38].

4.3. Main clustering results

Table 3 reports the six clustering metrics on all eight datasets. The analysis used 10,000 paired-bootstrap resamples per contrast and Holm correction across 24 contrasts separately for ACC and NMI. Relative to HLPNMF, ScaleSem-NMF achieves its largest stable gains on COIL100 (ΔACC 0.04546, 95% paired-bootstrap interval [0.04223, 0.04894]; ΔNMI 0.03041 [0.02847, 0.03251]); both Holm-adjusted sign-test values are below 5×10^{-5}. COIL20 also shows strong dual-metric gains. AR has positive paired-bootstrap intervals (ΔACC 0.00535 [0.00228, 0.00853]; ΔNMI 0.00246 [0.00132, 0.00359]), and MNIST improves over HLPNMF.

Figure 2 complements the mean-standard-deviation table with paired ACC and NMI differences and their uncertainty intervals for every dataset-baseline contrast.

ScaleSem-NMF produces positive paired ACC and NMI effects against all three baselines on AR and COIL100. COIL100 remains significant across all

Table 3: Clustering performance on eight datasets (mean% ± standard deviation%) over 20 paired seeds. Bold and underlining denote the best and second-best means, respectively.

Dataset	Method	ACC	NMI	Purity	F-score	Precision	Recall
YaleB	ScaleSem-NMF	**73.54 ± 1.43**	**73.77 ± 0.92**	**73.54 ± 1.43**	**57.26 ± 1.62**	**55.79 ± 1.71**	**58.81 ± 1.60**
	HLPNMF	73.25 ± 1.30	73.50 ± 0.82	73.25 ± 1.30	56.69 ± 1.50	55.11 ± 1.64	58.37 ± 1.45
	GOCNMF	73.22 ± 1.39	73.36 ± 1.01	73.23 ± 1.38	56.40 ± 1.66	55.00 ± 1.73	57.88 ± 1.67
	SNMFWLP	54.87 ± 2.81	63.03 ± 1.79	56.33 ± 2.57	21.83 ± 2.93	14.53 ± 2.62	45.09 ± 2.59
COIL20	ScaleSem-NMF	**97.33 ± 0.94**	**97.43 ± 0.65**	**97.33 ± 0.94**	**95.17 ± 1.48**	**94.67 ± 1.86**	**95.68 ± 1.19**
	HLPNMF	94.86 ± 1.28	94.88 ± 1.19	94.86 ± 1.28	90.73 ± 2.21	89.99 ± 2.43	91.47 ± 2.02
	GOCNMF	92.35 ± 1.20	92.06 ± 1.21	92.35 ± 1.20	86.34 ± 2.06	85.50 ± 2.31	87.20 ± 1.86
	SNMFWLP	67.93 ± 3.44	78.49 ± 3.18	71.16 ± 3.18	54.42 ± 5.25	46.51 ± 6.08	65.96 ± 4.31
Optdigits	ScaleSem-NMF	**98.56 ± 0.11**	**96.28 ± 0.24**	**98.56 ± 0.11**	**97.15 ± 0.20**	**97.14 ± 0.21**	**97.16 ± 0.20**
	HLPNMF	98.56 ± 0.14	96.26 ± 0.26	98.56 ± 0.14	97.14 ± 0.27	97.13 ± 0.28	97.15 ± 0.27
	GOCNMF	98.35 ± 0.13	95.83 ± 0.28	98.35 ± 0.13	96.73 ± 0.26	96.70 ± 0.26	96.75 ± 0.26
	SNMFWLP	93.15 ± 5.92	94.26 ± 1.71	94.07 ± 4.81	92.31 ± 4.97	89.42 ± 8.20	95.71 ± 1.50
PIE	ScaleSem-NMF	95.68 ± 1.18	**97.12 ± 0.51**	95.71 ± 1.14	92.43 ± 1.70	91.56 ± 2.08	93.33 ± 1.32
	HLPNMF	**95.77 ± 1.16**	97.10 ± 0.55	**95.77 ± 1.16**	**92.53 ± 1.73**	**91.72 ± 2.09**	**93.36 ± 1.38**
	GOCNMF	93.80 ± 1.24	95.51 ± 0.72	93.80 ± 1.24	88.79 ± 1.99	87.74 ± 2.30	89.86 ± 1.67
	SNMFWLP	57.56 ± 2.60	76.95 ± 2.15	63.33 ± 2.53	33.51 ± 3.67	23.69 ± 3.30	57.72 ± 3.24
AR	ScaleSem-NMF	**64.75 ± 1.27**	**84.54 ± 0.44**	**66.07 ± 1.22**	**51.99 ± 1.09**	**46.64 ± 1.32**	**58.74 ± 1.06**
	HLPNMF	64.21 ± 1.18	84.30 ± 0.39	65.56 ± 1.18	50.79 ± 0.95	44.96 ± 1.33	58.39 ± 1.04
	GOCNMF	63.31 ± 1.04	84.02 ± 0.48	64.94 ± 1.07	49.43 ± 1.60	43.30 ± 2.06	57.64 ± 1.12
	SNMFWLP	46.77 ± 1.48	73.99 ± 1.00	51.01 ± 1.26	23.57 ± 1.72	16.39 ± 1.55	42.19 ± 1.32
MSRA25	ScaleSem-NMF	**99.43 ± 0.12**	**98.81 ± 0.21**	**99.43 ± 0.12**	**98.85 ± 0.25**	**98.83 ± 0.26**	**98.88 ± 0.25**
	HLPNMF	99.41 ± 0.12	98.77 ± 0.21	99.41 ± 0.12	98.81 ± 0.23	98.78 ± 0.24	98.83 ± 0.24
	GOCNMF	97.45 ± 0.88	95.85 ± 1.21	97.45 ± 0.88	95.10 ± 1.72	94.85 ± 1.93	95.35 ± 1.56
	SNMFWLP	83.48 ± 3.19	86.49 ± 2.02	85.10 ± 2.40	78.72 ± 3.47	75.79 ± 4.80	82.06 ± 3.64
COIL100	ScaleSem-NMF	**92.24 ± 0.39**	**95.57 ± 0.19**	**92.24 ± 0.39**	**87.91 ± 0.60**	**86.95 ± 0.72**	**88.89 ± 0.54**
	HLPNMF	87.69 ± 1.05	92.53 ± 0.55	87.70 ± 1.04	80.14 ± 1.62	78.58 ± 1.87	81.75 ± 1.37
	GOCNMF	82.51 ± 1.19	88.00 ± 0.83	82.52 ± 1.19	71.37 ± 1.91	70.14 ± 2.09	72.65 ± 1.74
	SNMFWLP	41.12 ± 1.75	62.65 ± 1.53	46.77 ± 1.38	17.04 ± 1.25	11.13 ± 0.95	36.46 ± 1.99
MNIST	ScaleSem-NMF	**93.64 ± 0.25**	85.93 ± 0.41	**93.64 ± 0.25**	**88.04 ± 0.44**	**87.89 ± 0.46**	**88.19 ± 0.43**
	HLPNMF	92.96 ± 0.29	85.11 ± 0.42	92.96 ± 0.29	86.87 ± 0.49	86.65 ± 0.53	87.10 ± 0.46
	GOCNMF	93.40 ± 0.25	85.51 ± 0.38	93.40 ± 0.25	87.57 ± 0.42	87.34 ± 0.47	87.80 ± 0.39
	SNMFWLP	93.52 ± 0.29	**85.93 ± 0.39**	93.52 ± 0.29	87.76 ± 0.50	87.53 ± 0.57	88.00 ± 0.45

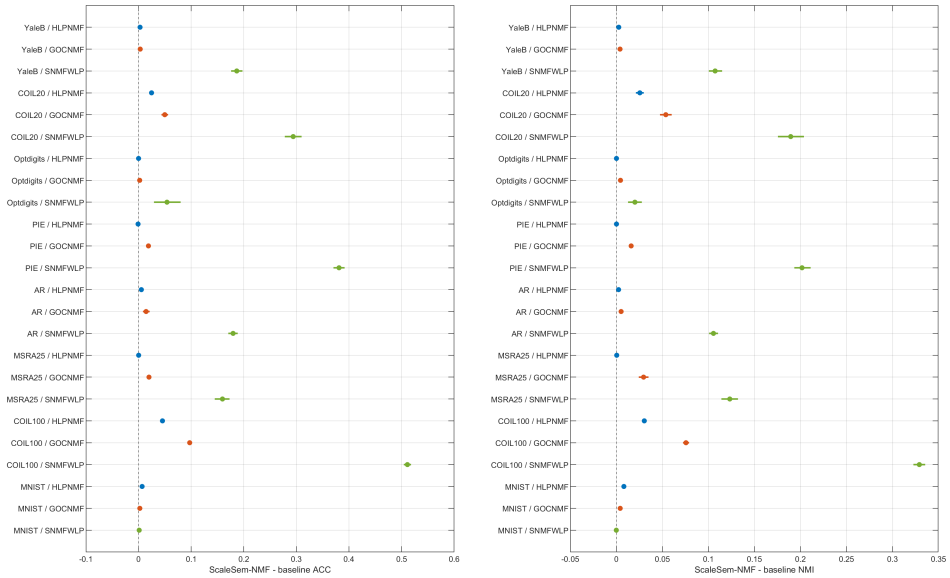


Figure 2: Paired ACC and NMI differences between ScaleSem-NMF and each baseline across eight datasets. Points denote mean differences, and bars denote 95% percentile-bootstrap intervals from 10,000 resamples.

24 Holm-adjusted sign tests, and AR has positive paired-bootstrap intervals.

4.4. Influence of supervision budget

The supervision experiment contains 4 datasets × 5 ratios × 20 seeds × 4 methods = 1600 runs, with the ACC and NMI trajectories shown in Fig. 3. At 5% supervision, ScaleSem-NMF delivers stable ACC and NMI on PIE, MSRA25, and MNIST and achieves the strongest performance on COIL20. The same pattern holds at 15%, demonstrating robustness across the tested supervision budgets.

4.5. Parameter sensitivity

The sensitivity matrix contains 255 runs over COIL100, COIL20, and PIE; Fig. 4 reports the corresponding ACC and NMI responses. With the

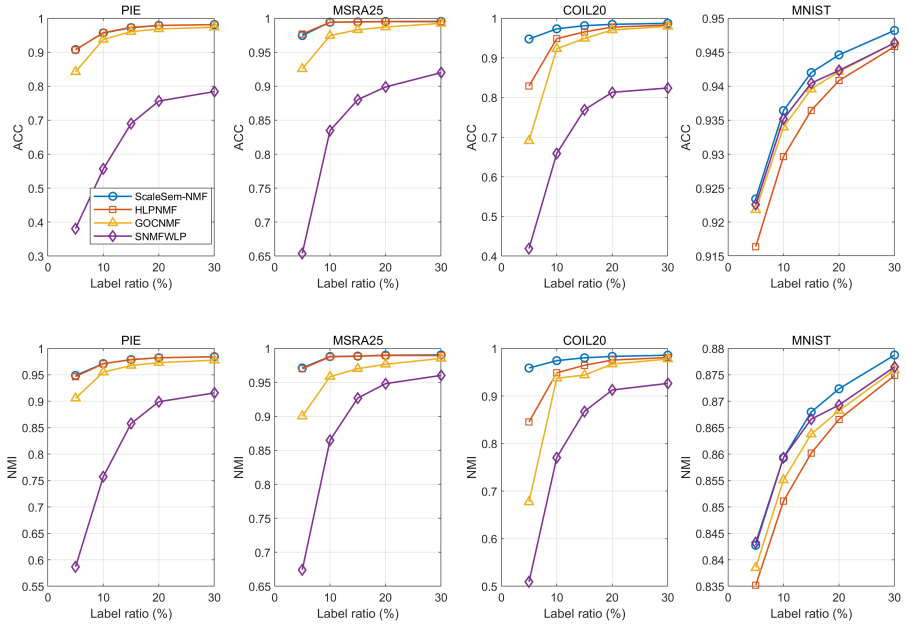


Figure 3: ACC and NMI across supervision budgets with the main-table parameters.

main-table parameters, the regularization weight α and propagation order k meet the local-stability threshold on all three datasets. Neighborhood size p shows the strongest sensitivity, with worst median ACC losses of 0.511, 0.448, and 0.148, respectively, making graph-neighborhood selection an important practical consideration.

4.6. Salt-and-pepper robustness

The noise matrix contains 400 unique runs. For salt-and-pepper corruption, the same corrupted images and pixels were shared across methods within each paired cell. Figure 5 shows stable ACC and NMI performance on COIL20, PIE, and YaleB at every corruption level from 10% through 30%.

At 30% corruption, ScaleSem-NMF exceeds the strongest baseline by 0.0291 ACC and 0.0203 NMI on COIL20; by 0.0107 ACC and 0.0067 NMI on PIE; and by 0.0089 ACC and 0.0009 NMI on YaleB. The corruption procedure

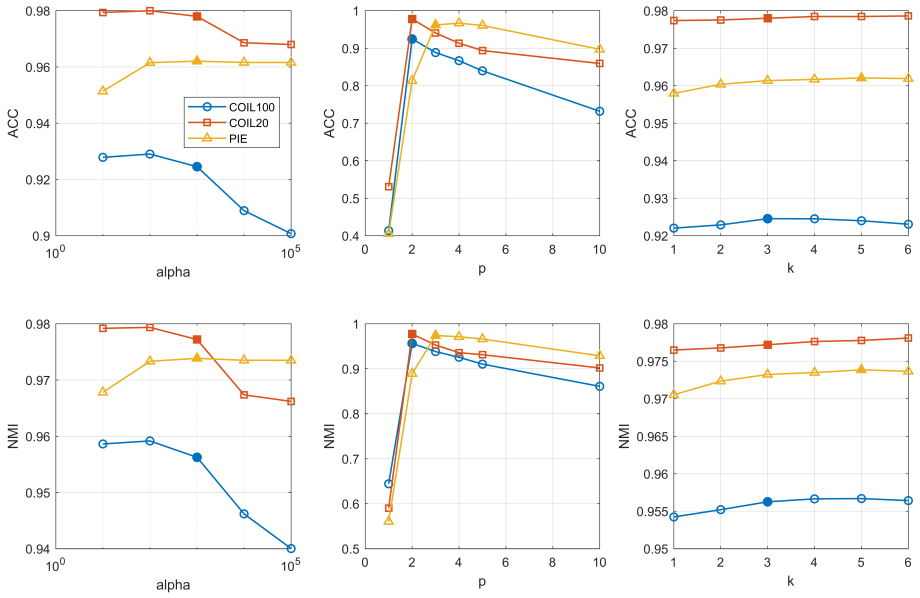


Figure 4: Parameter sensitivity of ScaleSem-NMF for ACC and NMI.

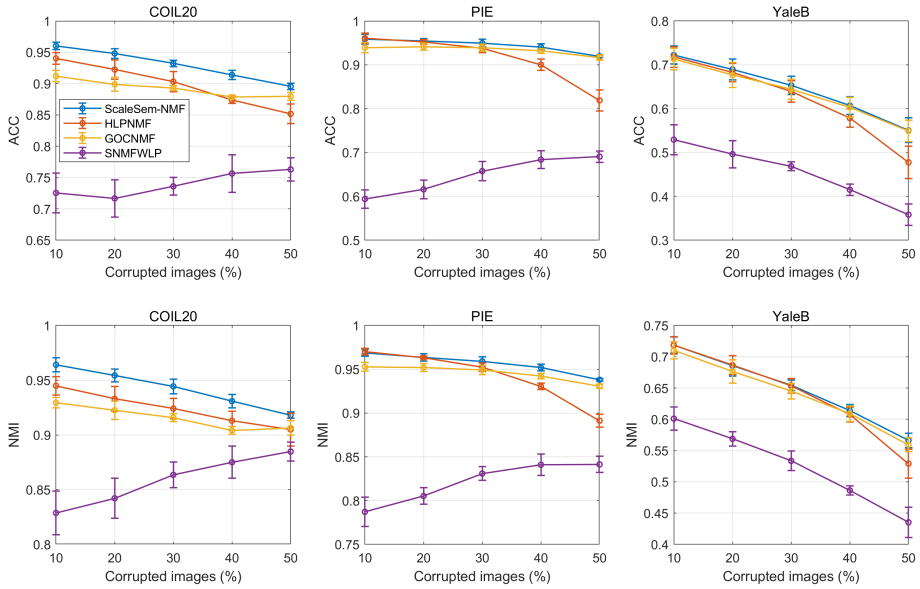


Figure 5: Salt-and-pepper robustness on three datasets.

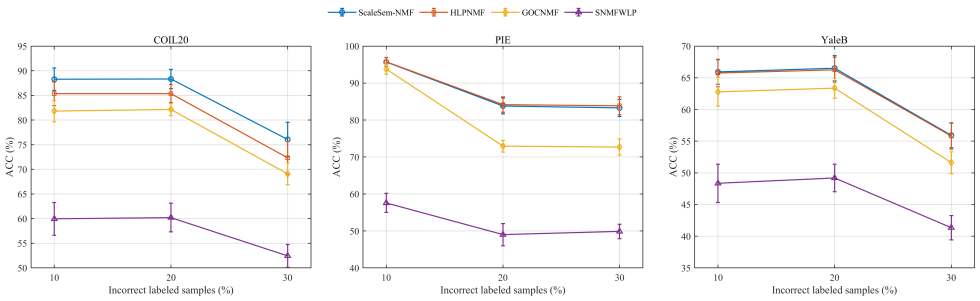
selects samples deterministically within each paired run and replaces exactly 10% of the 32-by-32 pixels in every selected image with balanced zero/255 impulses.

4.7. Incorrect-label robustness

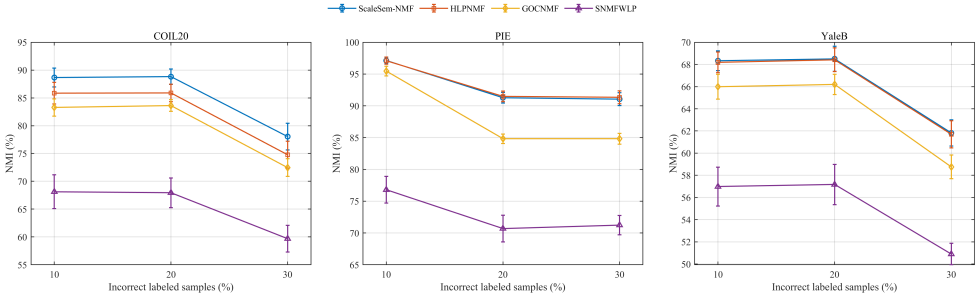
We further corrupted 10%, 20%, and 30% of the labeled samples within every class by assigning each selected sample a uniformly drawn incorrect class. The supervision budget, unlabeled evaluation set, model parameters, initialization convention, and native baseline decision rules remained unchanged. Figure 6 summarizes 720 runs over three datasets, three noise levels, four methods, and 20 paired repetitions.

ScaleSem-NMF has the highest mean ACC and NMI at all three noise levels on COIL20 and YaleB. On PIE, HLPNMF has higher mean ACC by 0.09–0.53 percentage points. ScaleSem-NMF has 0.02 percentage points higher NMI at 10% noise, and HLPNMF has 0.20 and 0.29 percentage points higher NMI at 20% and 30% noise. At 30% incorrect labels, ScaleSem-NMF obtains ACC/NMI of 76.06%/78.05% on COIL20, 83.30%/91.06% on PIE, and 55.93%/61.82% on YaleB.

Table 4 gives the paired ScaleSem-NMF–HLPNMF differences. All six COIL20 intervals are positive. On PIE, the ACC intervals and the NMI intervals at 20% and 30% favor HLPNMF, whereas the 10% NMI interval contains zero. On YaleB, the 20% ACC interval is positive and the other intervals contain zero.



(a) ACC



(b) NMI

Figure 6: Clustering performance under incorrect labels. Points and error bars denote the mean and standard deviation over 20 paired repetitions.

Table 4: Paired ScaleSem-NMF minus HLPNMF differences under incorrect labels (percentage points; 95% percentile-bootstrap intervals from 10,000 paired resamples).

Dataset	Incorrect labels	ΔACC [95% interval]	ΔNMI [95% interval]
COIL20	10%	$+2.93$ $[+2.46, +3.40]$	$+2.81$ $[+2.44, +3.20]$
	20%	$+2.97$ $[+2.53, +3.44]$	$+2.92$ $[+2.59, +3.27]$
	30%	$+3.74$ $[+3.19, +4.28]$	$+3.26$ $[+2.81, +3.72]$
PIE	10%	-0.09 $[-0.17, -0.01]$	$+0.02$ $[-0.06, +0.10]$
	20%	-0.36 $[-0.56, -0.16]$	-0.21 $[-0.29, -0.13]$
	30%	-0.54 $[-0.79, -0.27]$	-0.29 $[-0.43, -0.16]$
YaleB	10%	$+0.15$ $[-0.14, +0.44]$	$+0.15$ $[-0.01, +0.30]$
	20%	$+0.25$ $[+0.02, +0.48]$	$+0.09$ $[-0.03, +0.21]$
	30%	$+0.14$ $[-0.17, +0.46]$	$+0.12$ $[-0.04, +0.29]$

4.8. Unified single-thread runtime

The unified runtime evaluation comprised 160 runs across eight datasets, four methods, and five paired repetitions, all executed in MATLAB R2019a using a single computational thread. Figure 7 shows the runtime distributions, and Table 5 gives the paired medians. Timing includes graph construction, native optimization, and native decision, while excluding data loading, normalization, split generation, metric calculation, and file writing. The speed comparison uses a lower paired median on at least 6 of 8 datasets as its threshold.

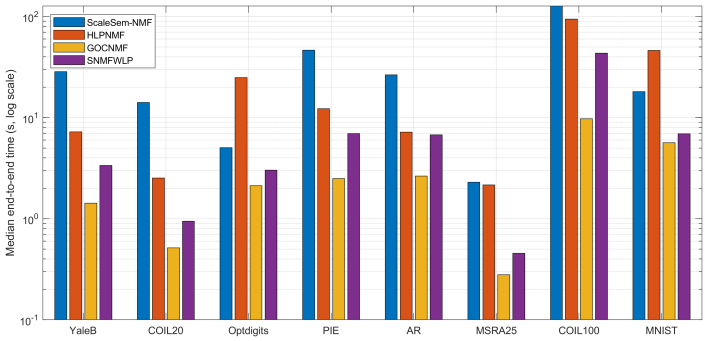


Figure 7: Unified single-thread runtime comparison.

Table 5: Median unified single-thread runtime in seconds over five paired seeds. Lower is better.

Dataset	ScaleSem-NMF	HLPNMF	GOCNMF	SNMFWLP
YaleB	28.510	7.221	**1.426**	3.364
COIL20	14.152	2.523	**0.513**	0.943
Optdigits	5.058	24.905	**2.122**	3.027
PIE	46.429	12.257	**2.493**	6.967
AR	26.544	7.185	**2.646**	6.742
MSRA25	2.295	2.163	**0.279**	0.454
COIL100	127.559	94.454	**9.771**	43.380
MNIST	18.082	46.196	**5.662**	6.896

ScaleSem-NMF is faster than HLPNMF on 2 of 8 datasets. GOCNMF

and SNMFWLP have lower runtimes across the tested datasets. These measurements quantify the computational profile associated with the observed clustering and salt-and-pepper results.

4.9. PIE scale and retained graph storage

Nested PIE subsets contain 10, 20, 40, and 68 classes with 420, 840, 1680, and 2856 samples. Figure 8 summarizes performance, runtime, and retained graph storage as the subset grows. All 20 runs are finite and monotone. Median runtime grows from 3.58 s to 43.96 s, while measured retained graph storage grows to 6.07 MiB on full PIE. Descriptive log-log slopes are 1.298 for runtime and 1.133 for retained graph storage over these four points. These measurements characterize computational feasibility through full PIE.

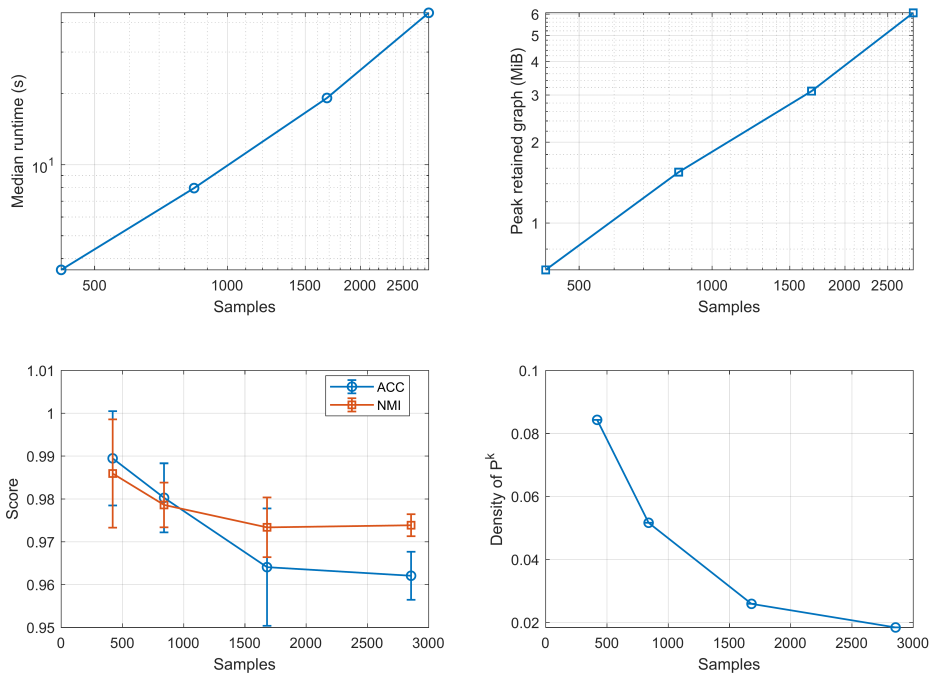


Figure 8: PIE scale, runtime, and retained graph storage through the full dataset.

4.10. Controlled mechanism ablations

Table 6 defines two controlled variants using matched splits, graphs, supervision budgets, parameters, initializations, and stopping rules. Their paired differences measure the contributions of adaptive amplitude and reconstruction coupling.

Table 6: Controlled ScaleSem-NMF ablations under matched evaluation and graph settings.

Variant	Sample amplitude s	Reconstruction contribution	Semantic propagation
Full ScaleSem-NMF	learned by exact nonnegative update	retained	retained
Fixed-scale	fixed for all samples	retained	retained
Propagation-only	not optimized (irrelevant)	removed	retained

Adaptive scale and reconstruction coupling both provide stable incremental value on COIL100, and the earlier protocol shows a strong effect on COIL20. Their effects are smaller on the remaining datasets, indicating that the contribution of each mechanism depends on dataset structure.

Figure 9 reports the paired ACC and NMI contributions of adaptive amplitude and reconstruction coupling. The largest mechanism effects occur on COIL20 and COIL100.

4.11. Cross-dataset mechanism diagnostics

To explain the cross-dataset variation in the ablation effects, we tested two hypotheses using the 20-seed outputs on eight datasets. The independent inferential unit is the dataset ($n = 8$). First, we tested whether the dataset-level coefficient of variation of learned amplitudes is positively associated with the ACC and NMI benefit of adaptive scale over fixed scale.

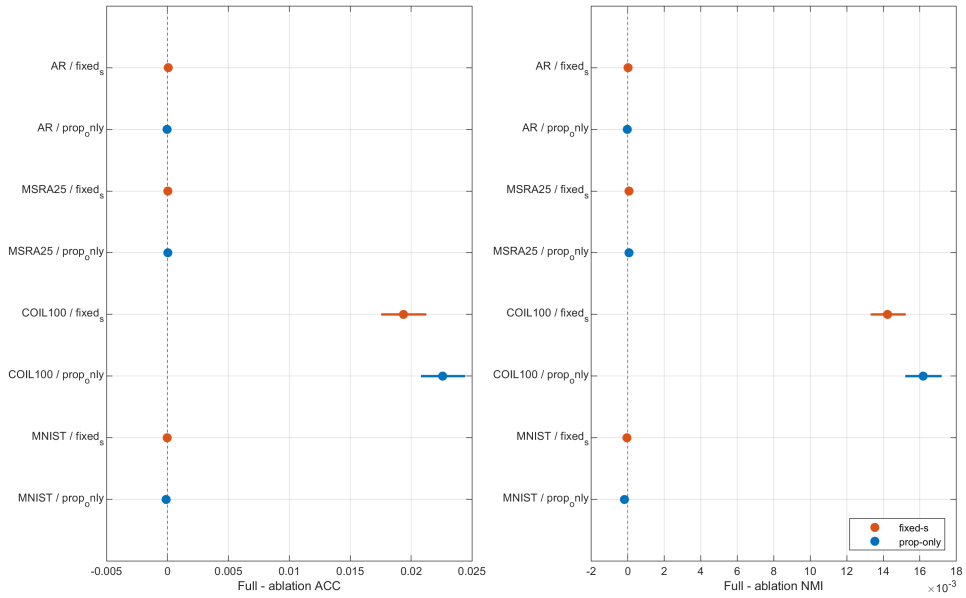


Figure 9: Controlled mechanism ablations with paired uncertainty intervals.

Second, we tested whether weaker alignment between reconstruction and propagation gradients is positively associated with the benefit of reconstruction coupling over propagation only. Exact one-sided Spearman permutation tests enumerate all $8! = 40320$ dataset permutations, and leave-one-dataset-out analysis checks directional stability. Table 7 reports the exact tests and their leave-one-dataset-out directional checks.

Table 7: Cross-dataset mechanism tests reporting effect direction, exact one-sided probability, and leave-one-dataset-out stability.

Predictor	Ablation gain	Spearman ρ	Exact p	Minimum leave-one-out ρ
Scale CV	fixed-scale ΔACC	0.5714	0.0756	0.5357
Scale CV	fixed-scale ΔNMI	0.5476	0.0855	0.5000
Negative gradient cosine	propagation-only ΔACC	0.1905	0.3323	0.0357
Negative gradient cosine	propagation-only ΔNMI	0.1905	0.3323	0.0357

Scale heterogeneity shows a positive association with adaptive-scale gains,

with exact probabilities and leave-one-dataset-out stability reported in Table 7. The negative cosine is below zero for every dataset, indicating aligned raw gradients, with values closer to zero representing weaker alignment. The empirical benefits are strongest on COIL20 and COIL100.

Figure 10 averages the simplex directions of the unlabeled samples in each true class. The diagonal masses are 0.879, 0.692, and 0.432 on COIL20, PIE, and YaleB, while the class-mean direction selects the corresponding semantic coordinate for 100%, 98.53%, and 100% of the classes. The learned amplitudes are also nondegenerate, with coefficients of variation 0.443, 0.568, and 1.138. Thus, the semantic and reconstruction-scale components exhibit distinct empirical structure.

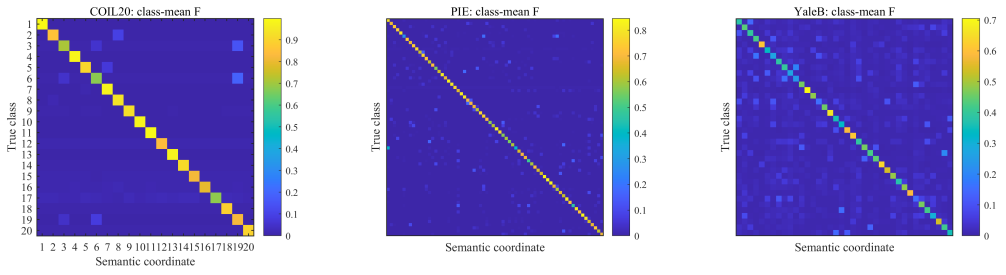


Figure 10: Class-mean semantic directions on unlabeled samples. Each row of a heatmap is the mean simplex direction for one true class.

The radial diagnostic in Fig. 11 directly compares the full model with its fixed-scale ablation on labeled samples. For each sample, the optimal nonnegative coefficient along the learned direction is

$$a_i^\star = \max\left\{0, \frac{(Uf_i)^\top x_i}{\|Uf_i\|_2^2}\right\}. \tag{14}$$

Across 20 paired runs, learning s_i reduces the mean labeled relative recon-

struction error on seven of eight datasets. The mean fixed-minus-learned reductions are 0.0330 on YaleB, 0.0026 on COIL20, 0.0041 on Optdigits, 0.0005 on AR, 0.0004 on MSRA25, 0.0011 on COIL100, and 0.0111 on MNIST; every corresponding 95% paired-bootstrap interval is positive. PIE changes in the opposite direction, with a mean difference of -0.0280 and interval $[-0.0316, -0.0244]$. The median within-run Spearman correlation between s_i and $a_i^\star$ is 1.000 on all eight datasets, numerically confirming the exact radial role of the amplitude update. This diagnostic isolates fitted radial reconstruction, while the controlled ablation in Fig. 9 quantifies its clustering effect.

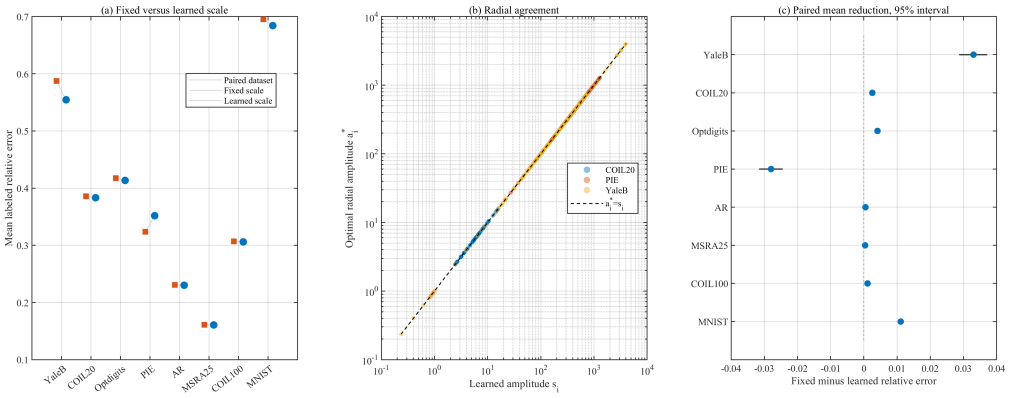


Figure 11: Radial-mechanism diagnostic over 20 paired runs. (a) Mean labeled relative reconstruction error for fixed and learned scale. (b) Learned amplitude and the optimal radial coefficient for representative datasets. (c) Mean paired reconstruction-error reduction with 95% percentile-bootstrap intervals.

Figure 12 compares the native coefficient representations of all four methods on the same class-stratified unlabeled subsets. The displayed projection uses a fixed random state. Across three fixed projection states, the mean 10-nearest-neighbor label purities of ScaleSem-NMF, HLPNMF, GOCNMF, and SNMFWLP are respectively 0.983, 0.953, 0.962, and 0.932 on COIL20; 0.904,

0.813, 0.830, and 0.823 on PIE; and 0.692, 0.580, 0.670, and 0.674 on YaleB. ScaleSem-NMF has the highest value in every dataset and projection state. The two-dimensional layouts and the repeated local-purity measurements jointly characterize the class geometry of the learned representations.

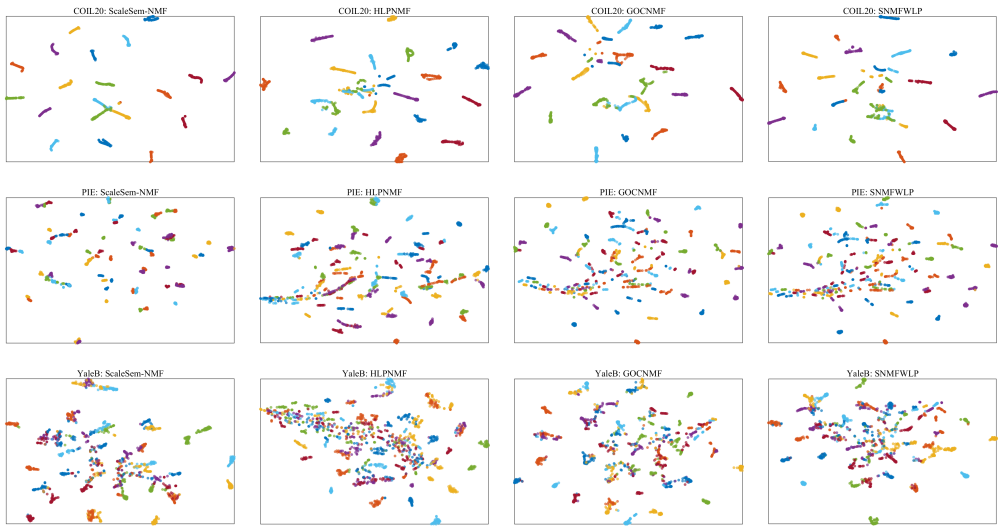


Figure 12: t-SNE projections of the native low-dimensional representations on matched unlabeled samples. Columns correspond to ScaleSem-NMF, HLPNMF, GOCNMF, and SNMFWLP; rows correspond to COIL20, PIE, and YaleB. Colors denote true classes for visualization.

On COIL20, the cosine similarities between the 20 learned basis columns and their corresponding normalized class means range from 0.982 to 1.000, with a mean of 0.996. The basis matrix U forms class-related reconstruction prototypes, while F supplies semantic direction and s supplies sample amplitude.

4.12. Convergence and implementation checks

Across 20 seeds on each of eight datasets, all objective histories in Fig. 13 are finite and no increase above tolerance is observed. YaleB, COIL20, PIE, and COIL100 reach the 200-iteration cap in every run and are reported as

cap-limited. Optdigits, AR, MSRA25, and MNIST have median iteration counts 66, 98.5, 49.5, and 58. Independent finite-difference checks of the three analytical gradients have maximum relative errors 9.20×10^{-7}, 5.64×10^{-7}, and 1.52×10^{-6} for the U, s, and F blocks, respectively; simplex, label, nonnegativity, graph-symmetry, and gauge-normalization checks also pass. These checks establish derivative accuracy, feasible iterates, and finite monotone descent, including in the cap-limited runs.

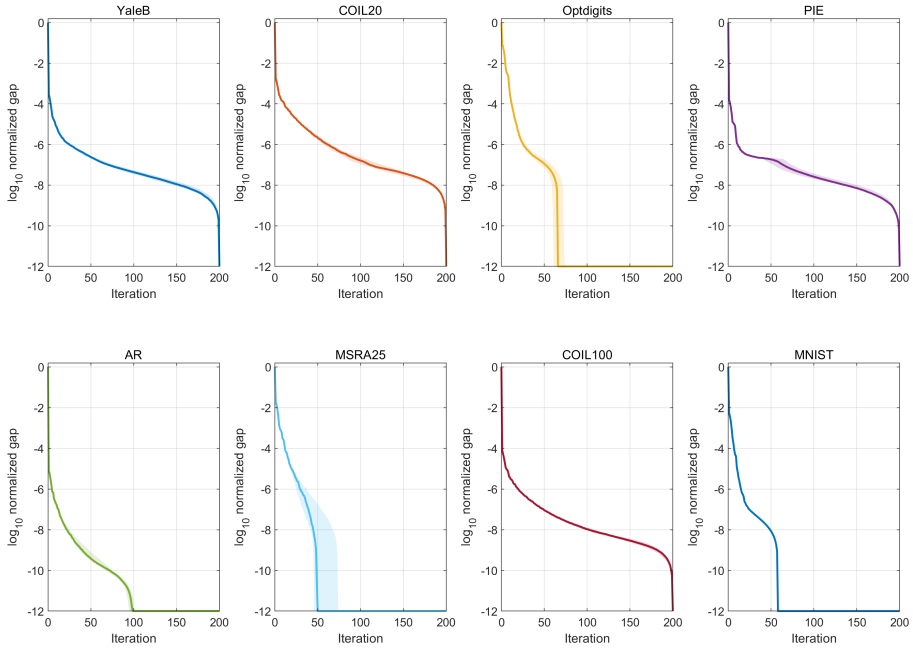


Figure 13: Objective histories on eight datasets. The curves show finite monotone descent; the text identifies the runs that reach the iteration cap.

5. Discussion

5.1. Empirical findings

Scale-semantic separation corrects the radial constraint imposed by point anchoring. COIL100 has the largest gains across baseline choice and statis-

tical summary, and COIL20 has the largest adaptive-scale ablation gains. The supervision-budget, salt-and-pepper, and incorrect-label experiments characterize performance beyond the clean 10% protocol.

5.2. Computational characteristics

The objective is nonconvex, and Theorem 2 states the regularity conditions for stationary accumulation points. Neighborhood size materially affects performance, and explicit high-order graph powers can become dense. Several datasets reach the iteration cap. Unified timing shows higher cost than GOCNMF and SNMFWLP across the tested datasets. The AR and selected external-baseline settings use preregistered domain-transfer parameters.

6. Conclusion

ScaleSem-NMF decomposes every latent coefficient into a learned reconstruction amplitude and a simplex semantic direction, thereby preventing a one-hot label from fixing an arbitrary radial magnitude. The model is compact and directly decodable. Eight-dataset experiments demonstrate strong dual-metric gains on COIL20 and COIL100 and positive effects on AR and MNIST relative to HLPNMF. Supervision-budget, salt-and-pepper, and incorrect-label experiments characterize limited-label and corrupted-data settings. Parameter and runtime experiments quantify neighborhood sensitivity and computational cost. ScaleSem-NMF provides an interpretable structural alternative for semi-supervised NMF.

CRediT authorship contribution statement

Deshu Sun: Writing-original draft, Methodology, Software, Conceptualization, Funding acquisition. Feng Wang: Writing-review and editing,

Visualization, Formal analysis. All authors reviewed the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

Data will be made available on request.

Declaration of use/non-use of Generative AI

The authors declare that no generative AI or AI-assisted technologies were used in the preparation of this manuscript.

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